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Modeling of barriers in integrating informal E-waste collectors into India's formal E-waste management system

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Abstract

Purpose – Despite the huge quantity of electronic waste generated in India, the collection and recycling rates in the country are remarkably low (0.93% as of 2016). The informal sector has a crucial role in managing India's e-waste. However, informal collectors direct the majority of the collected e-waste to informal recyclers leading to poorer resource recovery. Integrating informal e-waste collectors into the formal system could optimize e-waste value chains. The study aims to understand the challenges encountered by informal e-waste collectors when attempting to integrate into India's formal e-waste management system.

Design/methodology/approach – Insights were captured through a series of in-depth interviews with 23 informal e-waste collectors across high-density areas in Delhi such as Karol Bagh, Lajpat Nagar, etc. The qualitative data were thematically analyzed to identify critical barriers, which were then examined for their interconnections through a causal framework developed using System Dynamics Modeling.

Findings – The analysis of collected data revealed a range of codes that were organized under distinct subthemes, eventually culminating in three broad systemic barriers: Barriers related to government (such as lack of awareness about the existence of any policies and inconsistencies in the regulations), barriers faced by informal collectors in dealing with formal recyclers (such as lack of awareness on registered recyclers and competitive pricing challenges) and barriers intrinsic to informal collectors (such as limited operational capacity, primary focus on monetary gains and lack of awareness regarding the environmental consequences).

Originality/value – The study provides a novel perspective by focusing on the integration of informal e-waste collectors into the formal system through a causal framework, addressing a gap in the existing e-waste management literature in the Indian context.

Keywords E-waste management, Macro social marketing, Stakeholder theory, Sustainability, System dynamics modeling

Paper type Research paper

1. Introduction

E-waste management is a wicked problem that involves a multifaceted network of stakeholders and processes that range from waste generation to recycling. Various studies have highlighted the importance of the interconnectedness of micro and macro perspectives to devise solutions to complex societal problems known as wicked problems (Akaka *et al.*, 2021).

The concept of Macro Social Marketing (MSM) emerged due to the necessity to approach individual behavior change through a systemic lens (Kennedy, 2016) that may provide

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specialized solutions to combat wicked problems (Akaka *et al.*, 2021). Traditionally, MSM has been defined as the social change initiatives that are driven by governmental bodies (Wymer, 2011). However, contemporary definitions have explained this concept to include societal-level interventions using social marketing strategies that are led by influential entities within the social system such as Non-Governmental Organizations (NGOs) and other stakeholders (Domegan, 2008; Kennedy, 2016). The focus of MSM through a systemic lens is to foster collective behavioral change at multiple levels involving diverse stakeholders. A key area of interest here is understanding the barriers that may hinder or slow down behavior change in such social settings (Domegan *et al.*, 2020; McKenzie-Mohr and Shultz, 2014).

The global significance of issues arising from e-waste is evident as highlighted by Ikhlayel (2018) in his study. It poses potential threats to both human health and the environment due to inadequate waste management practices. As a result, the sustainable management of e-waste has evolved into a global need, signifying its paramount importance in the pursuit of the Sustainable Development Goals (SDGs) outlined in the 2030 agenda. This is particularly pertinent to Goal 3 ("Ensuring Clean Water and Sanitation"), Goal 8 ("Promoting Decent Work and Economic Growth"), Goal 11 ("Fostering Sustainable Cities and Communities") and Goal 12 ("Encouraging Responsible Consumption and Production"). Among all the sources of electronic waste, e-waste generated by households constitutes a significant share of 20%–21% (LeBel, 2012). Additionally, the collection of household e-waste presents notably greater challenges for formal collection enterprises compared to that originating from industries and institutions (Liu *et al.*, 2007). The current study is therefore focused on e-waste generated by households only.

Furthermore, India is the third-largest producer of e-waste globally with more than 3.23 million tonnes of electronic garbage generated annually (Forti *et al.*, 2020). Future forecasts suggest that by 2050, the nation's output in terms of e-waste will surpass 161 million tonnes (Biswas and Singh, 2020). Some studies also suggest that only a small percentage of India's overall electronic waste undergoes environmentally responsible recycling within the formal sector (Dutta and Goel, 2021; Biswas and Singh, 2020). The data released by the Ministry of Environment, Forests, and Climate Change shows that India has recycled only 32.9% of electronic waste generated in 2021–22.

The Government of India (GoI) created an advanced and well-thought-out plan to manage e-waste as a response to reduce the evil consequences on the environment and health hazards. The first major directive was outlined in 2011 as the E-waste (Management and Handling) Regulations that fell under the jurisdiction of the Environment Protection Act (EPA). The recent revisions were made to the regulations in 2018 and hence, the current E-waste Management Rules were subsequently formulated in November 2022. However, the practical implementation of these regulations has revealed a complex and challenging reality on the ground. One major hindrance is the existence of the informal sector that manages this waste stream extensively. This sector operates beyond the purview of current legislation and regulations. The current e-waste regulations regulate the formal recyclers and talk at length about the Extended Producer Responsibility (EPR) but ignore the informal sector's crucial role in the collection of e-waste.

There are well-established unorganized networks of garbage collectors/dealers in the country. Each of these entities functions on a modest scale, such as door-to-door rubbish collectors and dealers, referred to as "kabadiwalas" among the community (Sengupta *et al.*, 2022a, 2022b). Social class also influences their choice of occupation with kabadiwalas belonging to slightly higher but still marginalized backgrounds than waste pickers (Luthra *et al.*, 2023). Approximately 95% of all electronic waste produced is efficiently handled by this informal sector (Dutta and Goel, 2021; Borthakur, 2020). The informal e-waste collection in

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India thrives due to its substantial workforce, cost-effective collection methods, flexible work procedures and minimal operating expenses (Sengupta *et al.*, 2022a, 2022b). However, a major challenge is that these informal collectors direct the majority of collected e-waste to informal recyclers (Singh *et al.*, 2020).

While informal collection networks, in comparison, showcase higher efficiency than their formal counterparts, the formal treatment processes demonstrate greater capabilities in recovering resources from e-waste. Moreover, the recycling methods employed by the informal sector rely mainly on manual processes and thus, fail to consider any environmental or occupational health and safety measures.

Hence, to leverage the strengths of the strong and flexible network of informal e-waste collectors, it is crucial to integrate these kabadiwalas into the formal e-waste management system and guide them to align their operations with the e-waste regulations imposed by the government. This move has the potential to bring tremendous social, environmental and economic change (Sengupta *et al.*, 2022a, 2022b; Wang *et al.*, 2017). Therefore, the current study aims to understand the underlying dynamics that may affect the efficacy of the system by understanding the barriers encountered by informal e-waste collectors when attempting to integrate into India's formal e-waste management system. This research makes a theoretical contribution by extending the metatheoretical perspective in MSM by involving diverse and often conflicting stakeholders at all levels to cocreate potential solutions and shared future outcomes. It deepens the understanding of the dynamics of complex marketing systems and strengthens the ability to manage complex wicked problems. The study also introduces a new dimension of integrating the informal collectors into the formal e-waste management system, while allowing them their freedom to operate, yet governing their involvement with the system for the larger benefit.

2. Literature review

2.1 Contextual research setting

The literature on E-waste management shows a quantitative orientation with a strong focus on emerging markets. Multiple systematic reviews suggest that most of them rely on quantitative, statistical and modeling approaches (Andeobu *et al.*, 2021; Thukral and Singh, 2023). For instance, Alblooshi *et al.* (2022) conducted a bibliometric study on e-waste management studies in India analyzing various parameters quantitatively. However, multiple mixed method studies are also gaining prominence, particularly in the emerging markets (Nisha *et al.*, 2022). Some recent empirical research studies (see Table 1) in the context of e-waste management employs mixed methods across emerging markets such as Kenya, India etc. with some global focus on policy analyses.

2.2 Social marketing and macro social marketing

Social marketing traditionally involved the application of marketing principles to change individual behavior for the greater social good (Andreasen, 1994). However, the evolved idea of social marketing focuses on macro, micro and meso-level behavior change and defines the concept as application of marketing principles to promote behavior not only at the individual but at community level too (iSMA, ESMA and AASM, 2013). Hence, the field now embraces more participative approaches to involve stakeholders at multiple levels, foster partnerships and cocreate value for the individual as well as social benefits (Shaw *et al.*, 2024; Domegan *et al.*, 2020). This systemic approach brings together various stakeholders operating at multiple levels to form a partnership that can have a greater impact on the behavior change than the individualistic social marketing approach (Wymer, 2021). For instance, Shaw *et al.* (2024) advocated for a macro social marketing approach that

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Table 1. Literature review (recent empirical works)

Title	Citation	Method employed	Market
"Sustainable electronic waste management framework for academic institutions in India"	Valai Ganesh *et al.*, 2025	Quantitative	Emerging
"Policy pathways to sustainable E-waste management: a global review"	Grandhi *et al.*, 2024	Quantitative	Global
"Issues and solutions of electronic waste urban mining for circular economy transition: an Indian context"	Sharma *et al.*, 2021	Quantitative	Emerging
"Sustainable development assessment of household e-waste reverse supply chains from an environmental ethic perspective"	Yu *et al.*, 2025	Quantitative	Global
"Decoding e-waste challenges with hybrid AHP and ISM model approach: an initiative toward a cleaner future in India"	Singhdeo *et al.*, 2025	Quantitative	Emerging
"Waste not want not: a sequential social marketing pilot to reduce landfill destined e-waste"	Xi *et al.*, 2026	Mixed method	Developed
"Sustainable waste electrical and electronic equipment management guide in emerging economies context: a structural model approach"	Kumar *et al.*, 2022a, 2022b	Mixed method	Emerging
"Why is China struggling with waste classification? A stakeholder theory perspective"	Guo and Chen, 2022	Qualitative	Emerging
"behavioral reasoning theory (BRT) perspectives on E-waste recycling and management"	Dhir *et al.*, 2021	Quantitative	Developed
"An exploratory study on producer's perspective toward E-waste management: a case of emerging markets"	Thukral and Singh, 2023	Qualitative	Emerging
"The role of electronic waste management solutions and message framing in influencing consumer behaviors: Exploring the crowdfunding context."	Testa *et al.*, 2024	Quantitative	Global

increasingly involved multiple stakeholders in national social marketing interventions to enhance their effectiveness. In line with this, social marketing as a concept also duly acknowledges the significance of partnerships and stakeholder engagement in designing successful behavior change interventions (Kennedy, 2016; Hurley *et al.*, 2019; Hodgkins *et al.*, 2019). This recent collective behavior change approach to prioritize structural alterations and macro-level shifts and move beyond individual-focused adjustments involves MSM (Domegan, 2008; Kennedy *et al.*, 2018). It's noteworthy here to highlight that MSM doesn't

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reduce the importance of interactions at the micro (transactional) level. Instead, it aims to build connections across micro, meso and macro levels to overcome systemic shortcomings.

While macro-social marketers play a role in driving change, they are not the sole drivers, necessitating the engagement of various other groups in this process, emphasizing the need to involve stakeholders. The central focus is on understanding how behavior and exchange relationships affect the different stakeholders and groups involved in the system (Domegan *et al.*, 2020). Furthermore, several scholars have highlighted the need for deeper investigation into how diverse actors collaborate, negotiate and influence outcomes in social marketing systems (Duane and Domegan, 2019; Duane *et al.*, 2022). This gap underscores the importance of integrating stakeholder theory, especially in the context of developing nations where resource constraints and social inequalities might impact intervention outcomes.

2.3 Stakeholder theory

Stakeholder theory advocates for a broader perspective in the design, planning and execution of initiatives, extending the focus beyond immediate participants to encompass other influential actors involved in complex issues (May and Previte, 2016; Freeman, 1984). It offers a framework for addressing challenges from the perspectives of different parties involved, allowing effective targeting of essential aspects (Wang *et al.*, 2017; Du *et al.*, 2020).

In the context of e-waste management, it is crucial to analyze the roles of various stakeholders to enhance the understanding of how the e-waste management network operates and how it can be managed better. The processing of e-waste management consists of various steps; door-to-door e-waste collection, segregation, dismantling, refurbishing and formal/informal recycling (Kumar, 2021).

Door-to-door domestic e-waste collection is mostly conducted by informal collectors called kabadiwalas who are responsible to collect end of life materials such as newspapers, plastic, glass bottles, electronic devices and gadgets, etc. (Radulovic, 2017). These informal collectors most of the time also perform the operations of segregation and dismantling because it lets them sell different parts separately to maximize their profits (Dutta and Goel, 2021). When viewed from a behavioral lens, this action seems justified because the social class to which these collectors belong, struggles or operates to fulfill the lower level needs. Hence, their actions are influenced by the economic benefits associated with them (Yadav *et al.*, 2022).

The last step in e-waste management is recycling. The e-waste collection is processed to extract valuable materials that can be put to fresh use and the left-over is discarded. The recyclers fall into two categories; informal and formal. Informal recyclers recover various metals, plastic and all the waste generated during this process is disposed of inappropriately and contaminates the environment (Panwar and Ahmed, 2018). On the other hand, the formal recyclers are government-authorized entities that recycle and dispose of electronic waste using the best available technologies that are safe for the environment (Turaga *et al.*, 2019; Panwar *et al.*, 2018). These formal recyclers first use various mechanical methods to separate the metallic parts and then use metallurgical methods to recover precious elements from electronic waste (Rautela *et al.*, 2021; Arya and Kumar 2020). Like any other occupation, e-waste collection and recycling also poses hazards for the labor involved and for the environment (air, soil and water) (Jain *et al.*, 2023).

Combined with the lack of awareness about the potential health hazards (Ibifunmilola, 2024) and the economic motivation behind selling of their collection by informal collectors to informal recyclers, environment friendly recycling of the collected e-waste seems a far-

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fetched possibility. (Sengupta *et al.*, 2022a, 2022b). Similar challenges are reported in developing nations such as Bangladesh where informal collectors dominate collection due to cost efficiencies, while formal recyclers struggle with inadequate supply of electronic waste (Akter and Hossain, 2025).

To address these challenges, our research aims to analyze three primary stakeholders; the informal collectors themselves, the formal recyclers and the regulatory authorities. The study demonstrates that multi-party cooperation and coordination can serve as an effective solution by leveraging the interdependencies of the stakeholders.

One of the most popular methods for the identification and analysis of the characteristics of stakeholders to determine their importance is the Mitchell score method (Liu *et al.*, 2022). The characteristics of stakeholders are analyzed based on three major parameters: namely, power, legitimacy and urgency to determine their importance (Wood *et al.*, 2018) (Table 2). Power refers to the ability of a stakeholder to carry out their will within a social system without any resistance. Legitimacy is the perception that the actions of the stakeholder are desirable and appropriate according to the norms of the social system. The third parameter, urgency, is the degree to which a stakeholder would claim attention, which can be either because of the time sensitivity or criticality of the relationship. The key stakeholders in the present context that could be identified from the literature review include the government and regulatory authorities, formal recyclers, consumers who dispose of e-waste and informal kabadiwalas (Sengupta *et al.*, 2022a, 2022b).

Given these dynamics, stakeholder theory offers a valuable analytical foundation for identifying which actors matter most and why. However, to understand how the actions of the three stakeholders affect the integration of informal collectors effectively with India's e-waste management structure, there is a need for a complementary systems-based approach.

2.4 System dynamics modeling

Linear research often fails to explain the interdependencies across macro, meso and micro levels, hindering comprehensive comprehension of complex problems termed "wicked problems." Approaches like nonlinear causal modeling, specifically system dynamics modeling (SDM), are vital for understanding such complex issues (Domegan and Calis, 2017; Padela *et al.*, 2024). This approach is well suited to address the causal complexity and nonlinear interaction, arising due to behavioral factors among social systems by viewing them from multiple perspectives (Domegan and Calis, 2017). The central idea is that the feedback loops of a system are what drive its behavior (Sterman, 2000). This method as compared to the linear cause-effect chain approach offers a broader level of analysis by understanding the cyclical nature of mutual causality (Domegan and Calis, 2017). The current approach employing SDM also deals with the complexities of behavioral systems where interactions among elements can lead to unpredictable outcomes (Byrne and Callaghan, 2014).

Specifically, the use of SDM methodology is well-established to analyze the involvement of various stakeholders in a system. Combining the two methodologies enhances the understanding of the system, complementing the results with multiple stakeholder level implications and impacts (Elias *et al.*, 2001). For instance, a study by Pejic Bach *et al.* (2020) is based on the system dynamics analysis of sustainable urban development in China and provides a comprehensive analysis of the results for policy analysis and public outreach via environmental, social and economic impacts. SDM and stakeholder theory have also been effectively used together for analyzing sustainability performance of supply chains. Rebs *et al.* (2019) in his study on the simultaneous interplay of dynamic capabilities and sustainable supply chain management, discussed insights from the perspectives of multiple stakeholders.

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Table 2. Stakeholder analysis

Stakeholder	Power	Legitimacy	Urgency
Government and regulatory authorities	High power as they have the ability to set regulations, enforce compliance and allocate resources	High legitimacy due to their authority to enforce compliance to the regulations	Moderate urgency to formalize e-waste management system for a more sustainable environment
Formal recyclers	Moderate to high power because of their expertise, infrastructure and resources	High legitimacy as these are licensed organizations for e-waste recycling and conformance to regulations	High urgency because integrating with informal e-waste collectors can enhance their supply of e-waste and hence, improve efficiency
Informal E-waste collectors	Moderate power because of their informal status and lack of resources but very effective and flexible network	Low legitimacy due to their non-conformance to regulations and standards	High urgency because they are dependent on waste collection for their livelihoods but they need to supply e-waste to formal recyclers for sustainable e-waste management
Consumers	Moderate power because they can drive the demand of appropriate e-waste recycling and collection services	Moderate legitimacy since they contribute to e-waste and mostly do not dispose of e-waste as per the regulations	Low to moderate urgency because they prioritize convenience, affordability over environmental sustainability. However, consumer behavior is changing due to increasing awareness

Further, Silva and Sousa (2018) used stakeholder theory along all stages of the SDM in their study on system dynamics modeling of the Brazilian Space Program. These studies discuss at length (within their specific cases) how decisions or actions at one stakeholder level may impact (positively or negatively) the decisions of other stakeholders and influence their actions. This becomes a process of continuous evaluation (thus, a dynamic system) unless the desired level of performance is achieved.

3. Method

Systemic issues like e-waste management are complex, involving relationships and interconnections among stakeholders at various levels. Qualitative research methods are well-suited for examining systemic issues within social marketing (Samuel and Peattie, 2016) as they allow for in-depth exploration within real-world settings.

3.1 Data collection

The current study was conducted in the capital of India, Delhi. Delhi was chosen because it is one of the most densely populated cities in India and has various households using electronic and electrical devices. In the present research, purposive sampling method was used to select kabadiwalas operating in high density parts of Delhi (Karol Bagh, Lajpat Nagar, Mayur Vihar, Dwarka and Malviya Nagar). All the participants were informal kabadiwalas who collected various end-of-use articles such as newspapers, plastic bottles, glass bottles and electronic waste through door-to-door collection methods. The interviews continued till theoretical saturation was achieved, which occurred after 23 interviews when no new themes

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or insights emerged from successive transcripts (Bryant and Charmaz, 2007; Hennink *et al.*, 2017). The saturation was first observed at interview with P19. This was followed by four additional interviews to confirm the stability of the existing themes with no new insights observed. Table 3 presents the socio-demographic profile of the sample.

Both authors conducted semi-structured in-depth interviews. All interviews were conducted in person at the locations convenient to the respondents. These interviews were conducted by the authors themselves in Hindi, which is the national language of India, to ensure effective communication and understanding. Before obtaining their consent for participation, interviewees were guaranteed the confidentiality of their identities and were briefed on the research objectives. The ethical clearance for the study was obtained from the Institutional Ethics Committee of [Institute Name: XXX]. Each interview comprised two segments: initial inquiries regarding background, challenges and motivations related to e-waste collection, followed by questions regarding potential integration into the formal electronic waste management system. The participants were encouraged to share their perspectives, experiences and insights on various aspects of their work. The average duration of each interview was approximately one hour. It has been emphasized in a previous study that it is important to build rapport with the participants to obtain good-quality insights (Dundon and Ryan, 2010). Furthermore, follow-up clarification discussions were conducted with some participants to gain a nuanced understanding of their challenges, motivations and perspectives. While these follow-up conversations were not recorded, they provided deeper insights due to the established rapport with the participants.

3.2 Data coding and analysis

The current study employed thematic analysis followed by SDM to first identify the barriers (themes) from interviews and then model the causal dynamics of the system. Recorded interviews underwent *verbatim* transcription and translation by the first author to maintain the accuracy of the participants' responses. This step helped in maintaining transparency and also helped in enabling comprehensive analysis of the collected data. Furthermore, the transcribed and translated data was uploaded into QDA Miner Lite for systematic coding and analysis. The qualitative interviews were analyzed using Braun and Clark (2021)'s six phase thematic analysis framework. The rigor, transparency and validity were maintained by using criteria suggested by Lincoln and Guba (1985) and were implemented by employing peer debriefing, discussion and comparison (Padela *et al.*, 2024; Gupta *et al.*, 2020). The initial

Table 3. Sample characteristics

Subject	Sample *n* = no of respondents	23 *n*	 %
Gender	Male	23	100
	Female	0	0
Age	18–25	0	0
	26–40	11	47.8
	41–55	10	43.5
	Above 55	02	8.7
Income	Low	23	100
	Middle/ high	0	0
Education	Less than undergraduate degree	20	87
	Undergraduate	3	13.0
	Postgraduate and above	0	0

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coding process involved open coding of the data to explore themes. Subsequently, pivotal codes were selected based on contextual elements as suggested by Corbin and Strauss (2008). Furthermore, to ensure validity and reliability, these themes and the coding process were reviewed by the second author. These initial and pivotal codes were then systematically compiled into a codebook that included code definitions, inclusion–exclusion criteria and quotations. The first author engaged in discussion with the second author to brief them about the entire coding process, interpretation of themes and the complete data analysis approach. The second author gave feedback and critical insights on the coding process and themes to refine them further. This step was crucial in enhancing the trustworthiness and rigor of the data analysis process. The initial codes were focused on specific research questions leading to the targeted analysis of the relevant themes. Continuous iterations and comparisons of themes were conducted by both authors to identify the recurrent patterns and validate the identified themes. The identified themes were then triangulated against existing literature and secondary data that further polished and solidified the themes (Table 4). Throughout the analysis process, transparency was maintained by keeping detailed records of the entire coding process and the revisions conducted by authors.

Table 4. Themes

Themes	Subthemes	Codes	Respondents reporting this Sub-theme
Barriers w.r.t. Government	Unawareness of E-waste regulations	E-WASTE-RULES, GOV-POLICIES, LEGAL-UNDERSTANDING	P1, P3, P5, P7, P12, P18, P22, P14
	Lack of tax preference	NO TAX-PREFERENCE, GST-COMPLICATIONS, TAXATION OBLIGATION Vs. LEGAL STATUS	P2, P4, P9, P15, P20
	Lack of effective guidance and standards	GUIDANCE-STANDARDS, STANDARD-DEFICIENCY, LACK-GUIDANCE	P6, P8, P11, P14, P22, P1, P12
Barriers w.r.t. Recyclers	Lack of information on formal recyclers	FORMAL-RECYCLERS, LEGITIMATE-PROCESSORS, ACCREDITED-RECYCLERS	P1, P3, P5, P7, P10, P12, P13, P16, P19
	Competitive pricing challenges	COMPETITIVE-PRICING, PRICE-DIFFERENTIAL, PRICE-COMPETITION	P17, P21
Barriers w.r.t. Collectors	Small-scale operations	SMALL-SCALE, LACK-LARGE-SCALE, NO-INFRASTRUCTURE	P1, P23, P5, P7, P12, P18, P14
	Selective E-waste collection	PROFIT-DRIVEN, SELECTIVE-COLLECTION, MONETARY-PRIORITY	P3, P5, P7, P9
	Limited environmental awareness	ENV-AWARE, LOW-ENV-KNOWLEDGE, ENV-IMPACT-UNAWARE	P2, P4, P6, P7, P8, P12, P14, P18
	Ease of E-waste collection	EASY-COLLECTION, FLEXIBLE-COLLECTION, SIMPLE-COLLECTION	P10, P11, P13, P14, P15, P18, P19, P20, P22

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3.3 System dynamics modeling and causal model development

The current research brings to the limelight a managed system with three actors – the informal collectors, formal recyclers and the regulators. The presence of multiple actors creates an uncertain environment as it is difficult for each actor to accurately contemplate the outcomes of their actions which are also influenced by the simultaneous actions of other actors. Due to such dynamics, the system is characterized by behavioral issues, especially at the level of informal collectors, where profit is the prime motive (Davis and Garb, 2015). The presence of behavioral issues makes it difficult to study the system using common reductionist techniques that fail to capture the nonlinearity of behavioral outcomes. SDM overcomes these limitations. It is a suitable method to study behavior of complex managed systems in an uncertain environment (Padela *et al.*, 2024).

SDM is concerned with studying "WHY" various "MANAGED SYSTEMS" do not behave in the same way we wish or expect them to (Coyle, 2009). Studies within SDM begin with an in-depth understanding of the complex system under study leading to the development of a diagrammatic representation of the system. In SDM parlance, this is termed a Causal Loop Diagram (CLD) or influence diagram. Figure 1 shows the causal loop diagram that depicts the system studied in this research. It shows different variables that act simultaneously affecting the attainment of the system's primary goal of integrating the informal e-waste collectors in the formal system.

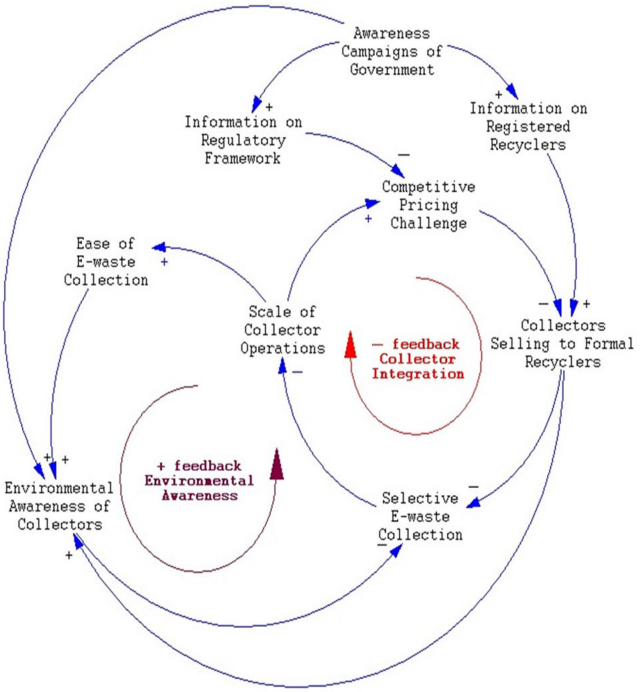


Figure 1. Causal diagram of interconnections between actor-related barrier subthemes

Source: Authors' own work

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The nonlinearity due to behavioral aspects involved in such complex systems is captured by the presence of feedback loops. The feedback loops are an essential component of a CLD. The presence of feedback loops emphasizes the need to provide feedback about the system's current performance so that it can be compared with the desired performance and a corrective action to deal with deviations may be taken. A positive feedback loop offers encouragement to the system to continue enhancing itself, while a negative feedback loop indicates the difference between expected and actual performance and hence, pushes the system to improve further aiming to reach the ideal level of performance characterized by no difference between expected and actual performance at any given point in time. The two types of loops act simultaneously to keep the system under balance. The polarity of a loop i.e. positive or negative nature, is associated with the polarity of the individual links comprising the loop. Every link (arrow) within the causal diagram is labeled with a "+" or "–" symbol, termed as the polarity of the link. Table 5 provides a summarized form of the polarity between different variables of the CLD. A positive polarity indicates that if the variable at the tail end of the arrow improves, the variable at the head end also improves. A negative polarity denotes an inverse movement between the variables at the tail and head ends of the arrow. In both scenarios, the variable at the tail end is the cause, and the variable at the head end is the effect.

Table 5. Cause and effect mapping summary

Loop	Cause	Effect	Link polarity	Net loop effect
Collector integration	Collectors selling to formal recyclers	Selective E-waste collection	-	-
	Selective E-waste collection	Scale of collector operations	-	
	Scale of collector operations	Competitive pricing challenge	+	
	Competitive pricing challenge	Collectors selling to formal recyclers	-	
Environmental awareness	Scale of collector operations	Ease of E-waste collection	+	+
	Ease of E-waste collection	Environmental awareness of collectors	+	
	Environmental awareness of collectors	Selective E-waste collection	-	
	Selective E-waste collection	Scale of collector operations	–	
Other links	*Cause*	*Effect*	*Link polarity*	*Net effect*
	Awareness campaigns of government	Environmental awareness of collectors	+	NA
	Awareness campaigns by government	Information on regulatory framework	+	
	Awareness campaigns by government	Information on registered recyclers	+	
	Information on registered recyclers	Collectors selling to formal recyclers	+	
	Information on regulatory framework	Competitive pricing challenge	-	
	Collectors selling to formal recyclers	Environmental awareness of collectors	+	

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SDM-based studies also involve quantitative simulation, necessitating the conversion of the CLD to a stock and flow diagram to simulate the model. Simulation of stock and flow diagrams helps in observing the results of specific actions taken by the actors in the system (Ammirato *et al.*, 2022). The need for simulation and to develop a stock and flow diagram is determined by the objectives of the research. The present research is conducted to suggest what could be the suitable policies that would lead to the integration of informal collectors into the formal E-waste management system. At this stage, an understanding of the various challenges involved should suffice, as the policymakers are required to understand who is going to be affected by their decisions. Hence, this purpose is effectively achieved with the CLD in Figure 1. Development of stock and flow diagrams would be more appropriate where results of specific policies on the prime variable of interest - number of informal collectors selling to formal recyclers - was to be observed, under the influence of other forces (variables) in action at the same time (Vanderby and Carter, 2010).

4. Results

4.1 Thematic analysis

The core focus of the present research is to employ a systems view to understand the barriers faced by the informal e-waste collectors that may hinder their integration into the formal e-waste management system. These barriers have been analyzed from the perspective of the informal collectors, as they interact with other stakeholders in the system such as regulatory bodies and formal recyclers. It is imperative to acknowledge and address diverse challenges from the perspectives of collectors when dealing with the system to foster the sustainability of such partnerships and ensure the long-term viability of this integration. While the current study relied on interviews conducted with informal e-waste collectors, the findings revealed that these challenges faced by collectors are part of a broader system. The analysis of collected data revealed a range of codes that were organized under distinct subthemes, eventually culminating in three broad systemic barriers: barriers related to government, barriers faced by informal collectors in dealing with formal recyclers and barriers intrinsic to informal collectors. These barriers faced by the informal collectors in their interactions with the stakeholders are detailed in Table 4.

4.2 Barriers faced by informal collectors in dealing with formal recyclers

The results indicate that informal e-waste collectors face various challenges when dealing with formal recyclers. The formal recyclers possess high urgency because of the need for a continuous supply of e-waste but moderate power over informal collectors that gives rise to this theme. Due to this imbalance, the primary barriers faced by them are a lack of awareness of registered recyclers and challenges related to competitive pricing. P17 emphasized in his interview that one of the reasons they don't give their e-waste to formal recyclers is that they don't pay as well as the informal ones. These barriers not only indicate the challenges faced by the collectors but also highlight the points of weakness in the overall formal e-waste management system. Formal recyclers often struggle to acquire waste at competitive rates from informal kabadiwalas. One possible reason is that registered recyclers are required to adhere to environmental, health and safety regulations. These regulatory demands imposed on the formal recyclers inflate their costs putting them at disadvantage as compared to informal recyclers who are able to offer better rates to the informal collectors for their e-waste collection (Turaga *et al.*, 2019). This finding is in alignment with findings from two studies; one conducted by Liu *et al.* (2022) in China and another by de Oliveira Neto *et al.* (2025) in Brazil. Addressing this gap may prove challenging without support from producers or the government. This systemic pricing imbalance poses a challenge for formal recyclers

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who are competing with informal ones and hinders the integration of informal collectors into the formal system. It is important to note here that formalizing informal recyclers, which might seem like an obvious solution to this challenge, is hindered by various constraints such as limitations of finances, and human resources.

The findings also revealed the lack of awareness among e-waste collectors about registered recyclers and their details. P3 mentions that he doesn't know about any registered recyclers and has no idea about the regulations either. There is a knowledge gap within the e-waste collection sector that hampers their ability to collaborate and send collected e-waste to formal recyclers. Kabadiwalas generally have limited understanding or awareness of any registered recyclers. This lack of information about the existence of formal recyclers is a systemic barrier that may make it difficult for informal collectors to engage and build relationships with formal recyclers.

4.3 Barriers faced by informal collectors concerning regulatory bodies

Systemic barriers faced by the informal collectors concerning government policies and regulations make it difficult for them to integrate seamlessly with the formal established system. These barriers emerged due to high legitimacy, high power but moderate urgency possessed by this group of stakeholders. The imbalance leads to inconsistency in regulations, limited policy awareness, and low communication. P1 mentioned in his interview:

> I am not aware of any regulations related to e-waste collection or transport. Also, I have no idea that there exists a formalization or registration process for the collectors of e-waste. No one has informed me about it.

The findings of the present study highlight two related yet distinct challenges faced by the informal collectors concerning the regulations. The first challenge is the lack of awareness about the existence of any policies and regulations surrounding e-waste management. During the interviews, many informal collectors mentioned that they had no idea about any regulations they have to follow while engaging in e-waste collection, trade or dismantling. This lack of guidance and communication from the regulatory agencies pose a significant barrier.

Another challenge is the inconsistencies in government policies, particularly the absence of any tax preference provided by the government. Although the current taxation system does not acknowledge informal collectors as legal entities, a 5% GST is levied on the trade and transport of electronic waste. The situation becomes very ironic when the system and regulations charge tax on the movement of electronic waste but neglect the final destination of these goods (Turaga *et al.*, 2019). This creates inconsistencies and confusion that further complicates the effort to integrate informal collectors into the formal system. A similar study conducted in Ghana reveals similar policy inconsistencies. The import and export of e-waste happens informally without being fully tracked but the support for the informal sector remains limited (Finn *et al.*, 2025).

The findings of the current study also reveal that the collectors feel they have no guidance from the government regarding the regulations surrounding electronic waste, the formalization process or even the requirement of formalizing themselves to legally trade e-waste. This lack of guidance is a major challenge that is hindering these informal, small-scale collectors from engaging with formal recyclers. For instance, some informal collectors mentioned (P1, P3, P12) in the interview that they are aware that some regulations exist related to e-waste, however, they are clueless about the specifics of these regulations and the formalization process.

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4.4 Barriers intrinsic to collectors

In addition to the external barriers, the informal collectors also face some internal challenges that hinder their ability to integrate into the formal e-waste management system. These barriers stem from low legitimacy, moderate power and high urgency of this group of stakeholders. This imbalance leads to barriers, including limited operational capacity, primary focus on monetary gains and lack of awareness regarding the environmental consequences. Most of these collectors have a very limited infrastructure and resources and hence, their scale of operation is also very small. During interviews, various collectors expressed concerns about their ability to engage with registered large-scale recyclers, citing their perceived insignificance in comparison. P18 said "we are too small to collaborate with big recyclers. Even if we wanted to, we do not have money or space to grow." Hence, this small scale of operation becomes a perceived barrier wherein they feel they are too small to integrate into the larger formal system. Even if they want to expand their operations to meet the standards that are set by the registered recyclers, they are not able to do that due to their limited infrastructure. Additionally, they feel that increasing the scale of operations might hamper their flexibility which is a major asset that sets them apart from the formal e-waste collectors. Hence, they continue to operate on a smaller scale that offers them more flexibility and autonomy but widens the gap between informal and formal systems.

Moreover, the primary focus of most of these collectors is financial gain and hence, they only focus on collecting the electronic waste that gives them more profits. They tend to reject items that will not fetch them a good price or might even dispose of those with regular solid waste. P7 explained that "he only collects the items that can be sold at a good price, such as specific metals or electronics that can be refurbished or contain more valuable components." This finding highlights a prevailing misconception that only high-value IT products are considered electronic waste while lower-value consumer products such as CRT Televisions or lamps are not classified as e-waste (Turaga *et al.*, 2019). This selective behavior of informal collectors leads to inappropriate disposal of lower-value electronic waste items.

The findings also highlight the lack of awareness among these collectors about the negative impact of inappropriate disposal or recycling of e-waste on the environment which is in line with Turaga *et al.* (2019). P 14 stated in his interview "we just throw in the dustbin the parts that won't sell. We never knew it will be harmful to the environment." This unawareness leads them to overlook the negative consequences of informal treatment processes on the environment creating an additional systemic challenge that may inhibit them from engaging with the formal recyclers.

Another interesting finding of the study is that these collectors have a very strong and flexible network that helps them collect e-waste much more efficiently than the formal collectors. It is not a challenge for the kabadiwalas but this asset does pose a challenge for the formal collectors who struggle to compete with them (Turaga *et al.*, 2019). This finding highlights a competitive advantage of the informal network of the collectors that further complicates the efforts to create a more unified formal e-waste management system.

4.5 System dynamics modeling: causal loop diagrams

The barriers (subthemes in Table 4) identified through thematic analysis together constitute the system level factors that affect the integration of informal e-waste collectors with the formal e-waste management system. The simultaneous interplay of these factors is captured in the proposed CLD of Figure 1 that has been developed in parlance with SDM. Figure 1 shows nine variables distributed among two loops governing the system's behavior. These are Collector Integration Loop and Environmental Awareness Loop. The challenges in integrating the informal collectors into the formal system arise at the ends of the government,

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recyclers and the informal collectors themselves. The governmental challenges "unawareness of e-waste regulations" and "lack of effective guidance and standards" are captured by the variable "Awareness Campaigns of the Government." This variable impacts other CLD variables "Information on Regulatory Framework" which corresponds to the governmental challenge of "Lack of Tax Preferences" and "Information on Registered Recyclers" that corresponds to the first challenge with respect to recyclers in Table 4. Further, remaining subthemes in Table 4 are captured almost *verbatim* as variables in the CLD. These include "Competitive Pricing Challenge" from the recyclers and "Scale of Collector Operations," "Selective E-waste Collection," "Environmental Awareness of Collectors" and "Ease of E-waste Collection" at the end of informal collectors.

4.5.1 Goal-seeking negative feedback loop: collector integration. A negative feedback loop depicts the gap between the actual and benchmarked/expected performance of a system. The prime objective of the system proposed in Figure 1 is to bring as many informal e-waste collectors (kabadiwalas) into the formal system where they sell their collection to formal recyclers recognized by the government. Hence, the negative feedback loop "Collector Integration" starts with the variable "Collectors Selling to Formal Recyclers." The success of the proposed system is depicted by a consistent increase in this number, eventually leading to all informal collectors in the country selling their e-waste collection to formal recyclers. Currently, the informal collectors are selective in the e-waste that they collect from households. If the informal collectors sell their collection to the registered recyclers at government-approved rates resulting in better profits for them, it will lead to a decrease in the Selective E-waste Collection (SEC) by them, because registered recyclers work on a wide variety of e-waste. Therefore, higher the number of collectors selling to registered recyclers, lower will be the problem of selective e-waste collection, due to enhanced understanding and knowledge about the variety of e-waste that can be handled. Hence, the relation is depicted with a negative polarity. As the selective collection of e-waste by collectors goes down, it results in an increase in the Scale of Collector Operations as these collectors are now involved in collecting a larger variety of e-waste and such collectors become the preferred collectors by a larger number of households willing to dispose of their e-waste responsibly. The opposite direction of movement of the two variables is depicted by a negative sign. An increase in the scale of collector operations gives them an increased negotiating power for their collection, thereby increasing the prices at which they sell to the formal recyclers. This will increase the cost burden on formal recyclers, thereby increasing the Competitive Pricing Challenge faced by them. Since the challenge of competitive pricing moves in the same direction as the scale of collector operations, the causal relationship between them is depicted with a positive polarity. Due to competitive pricing, the recyclers offer lower prices to the collectors which forces the collectors to return to non-registered aggregators/recyclers that offer higher prices. This ultimately brings down the number of collectors selling to registered recyclers, increasing the gap and defeating the main objective of the system to integrate as many informal collectors as possible. This pushes the system to adopt policies and practices to reduce the gap. This loop is a negative feedback loop. A multiplication of 3 negative and 1 positive polarity results in overall negative polarity for the loop.

Working with formal recyclers also increases the environmental awareness of the informal collectors, which further brings down the selective collection practice by kabadiwalas. This relationship is outside the Collector Integration Loop but affects the loop variable selective e-waste collection.

4.5.2 Growth producing positive feedback loop: environmental awareness loop. The positive feedback loop involves four variables – Scale of Collector Operations, Ease of E-waste Collection, Environmental Awareness of Collectors and Selective E-waste Collection.

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The collector integration loop incorporates the cause-and-effect relation between Selective E-waste Collection and the Scale of Operations. The Scale of Operations increases as Selective E-waste collection goes down. In the Environmental Awareness Loop, increasing the scale of operations of the collectors makes them reach out to a larger number of households and other set-ups thereby increasing the variety of e-waste that they collect. Such kabadiwalas become the preferred e-waste collectors of households for all kinds of waste. Due to this, the collectors do not have to head-hunt too hard for the e-waste as the customers reach out to them directly, enhancing the Ease of E-waste Collection for informal collectors. An increased reach to multiple households as sources of e-waste further enhances the knowledge of the collectors about the different kinds of e-waste that can exist. This way, their knowledge about the variety of e-waste items is enhanced, which further brings down the practice of selective collection by them. An increased environmental awareness of the collectors motivates the collectors to collect even such electronic waste which may not be recyclable or reusable, still requires careful dumping, therefore further bringing down the practice of selective collection. The presence of two positive and two negative polarities, give an overall positive polarity to the loop.

Although not a part of the loop, the direct effect of Collectors Selling to Formal Recyclers and Awareness Campaigns by the Government on Environmental Awareness of Collectors needs a mention. Both variables impact the environmental awareness of collectors via a positive polarity. Environmental awareness includes an appreciation of the need to handle electronic waste safely, the role and importance of collectors in the e-waste value chain, what items are classified as electronic waste, which recyclers to sell and the government-approved rates of the collected electronic waste. The variables ease of collection, awareness campaigns of the government and selling to formal recyclers, together work to enhance the awareness of collectors regarding proper handling and disposal of electronic waste.

Both the loops – collector integration and environmental awareness – are primarily dependent on initiatives by the government to influence the behavior of various stakeholders appropriately. For instance, the government may consider running informative campaigns for the informal collectors in different localities, highlighting the benefits of integrating with the formal system. Such campaigns may also aim at communicating the penalties for not following certain practices at both collectors' and recyclers' levels. However, at the current stage, introducing negative penalties may deter the collectors from integrating with the formal e-waste recycling system and may possibly lead to few recyclers adhering to the minimum requirements to avoid penalties rather than working to make the system more efficient.

4.5.3 Exogenous variables, intermittent variables and resulting connections. Apart from the variables that fall in one or more loops, there are few exogenous variables. Exogenous variables are those that are not under direct control of the system but they still impact the desired results of policies and practices adopted by the system. In a causal diagram, an exogenous variable can have only outgoing arrows and no incoming arrows. The exogenous variable in the CLD of Figure 1 is Awareness Campaigns run by the Government. Such campaigns aim to increase awareness among collectors, recyclers and the public in general about the adverse environmental effects of e-waste and the importance of its responsible handling and disposal. However, in the context of this research, we focus primarily on the informal collectors as the intended audience of these government-run campaigns. The collectors not only collect e-waste but they are also producers of e-waste just like any other individual. The awareness campaigns guide the collectors in recognizing e-waste and other hazardous materials among regular household waste (generated in their own

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households and collected from other households) and encouraging them to sell to formal recyclers registered with the government.

5. Theoretical implications

Effectively managing e-waste, especially during its collection stage, has emerged as a major challenge, impeding its appropriate treatment and recycling processes. This study systematically examines the obstacles faced by informal e-waste collectors in India as they interact with diverse stakeholders. Using SDM, it delves into the interconnections among these barriers.

The findings of this research offer valuable contributions to both theoretical and practical aspects of macro-social marketing. From a theoretical perspective, the current study contributes by providing a metatheoretical perspective for understanding the dynamics of marketing systems especially in solving wicked problems like e-waste management. It is crucial to devise strategies and solutions surrounding the behavioral and structural dynamics to address the relational aspects of the marketing systems. The study employs SDM technique to provide insights into the interconnectedness and relationships of the barriers faced by informal e-waste collectors while integrating into the formal e-waste management system. Qualitative causal models play a crucial role in uncovering the direct as well as indirect impact and the unintended outcomes of actions of various stakeholders who may have conflicting interests (Stepp *et al.*, 2009; Padela *et al.*, 2024).

6. Practical/policy implications

From a practical perspective, the current study contributes largely toward framing of policies that may facilitate the integration of informal e-waste collectors into the formal e-waste management system. The findings emphasize the need to design and implement more holistic, stakeholder-aligned interventions rather than relying on surface-level interventions to manage e-waste effectively. The three categories of barriers identified in the study reveal that each stakeholder group operates with distinct constraints and motivations, with the profit motive being common among informal collectors and formal recyclers. The interventions should hence address these differentials through mutually reinforcing and coordinated strategies (Frey, 2000).

Important observations emerge from the causal model in the current study. One is that the system will remain ineffective in the absence of awareness campaigns targeted toward informal collectors and formal e-waste recyclers. This is evident from the exogenous characteristic of the variable Awareness Campaigns by the Government in the CLD. The variable impacts one or more loop variables either directly or via other variables, indicating that the system may collapse if this variable is absent. Further, the findings also highlight that informal collectors find the current regulations and policies as difficult and inaccessible. A government-led campaign must be launched that focuses on bridging this gap by providing not only awareness but also training to the collectors. These training programs can aim to educate the informal collectors about the formal recyclers in their area, the regulations and standards surrounding e-waste collection, recycling and dismantling and the overall formalization process. In addition, some support can be provided to these informal collectors in understanding the current regulatory framework. It might aid in better monitoring of the e-waste flow to ensure formal recycling of the e-waste collected by the informal e-waste collectors. The awareness campaign may also educate the collectors about the negative consequences of the informal treatment process and incorrect disposal of e-waste on the environment. By integrating the awareness initiative with monetary rewards, a more

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equitable strategy could be attained. The campaigns may also emphasize the nonperformance parameters for both collectors and recyclers and the associated penalties.

Two, the informal collectors have a very big advantage in the form of their reach to individual households and a very strong and flexible network that cannot be replaced. In addition, any move trying to formalize these collectors might hamper their flexibility and adaptability. The policymakers can devise policies that may allow these collectors to continue collecting e-waste from households with minimum interference. The current regulatory framework should be tweaked to offer some concessions and tax preferences to the informal e-waste collectors that may permit their door-to-door collection with minimum intervention while maintaining some transparency and control. Therefore, a government-led macro-social marketing campaign like Swachh Bharat Mission (Clean India Mission) could be pivotal in reshaping consumer behavior and perceptions around e-waste management. The informal e-waste collectors currently function in isolation from refurbishers, dismantlers and recyclers. The interventions should follow a systems-thinking approach to design vertically integrated mechanisms by creating shared logistical arrangements and offering skills development pathways to the collectors.

Finally, one of the key findings of the current study is the competitive pricing challenge faced by formal recyclers. Despite numerous campaigns, the pricing challenge may still hinder the integration of informal collectors into the formal system as the former may keep selling the collected e-waste to informal recyclers for more money. Future policy interventions may focus on employing strategies that may balance this competitive pricing challenge in the market with long-term environmental sustainability. This may include incentivizing the formal recycling processes or providing subsidies to the formal recyclers that may help bridge the price gap. The limited engagement of formal recyclers with informal collectors highlights the need for structured collaboration through transparent pricing, guaranteed buy-back arrangements and shared collection centers. Since formal recyclers possess high legitimacy, such partnerships can enhance system efficiency, reduce conflicts and build legitimacy for informal actors without requiring abrupt formalization.

This research proposes that informal collectors should be integrated into the formal recycling system, while allowing them their freedom to operate. It will result in benefits during all the stages of the Life Cycle of Electrical and Electronic Equipments (EEE), with greatest impact during the third stage "pre-processing and re-processing" (Kumar, 2021). Figure 2 depicts this integration of informal collectors with the EEE life cycle. The black dot represents "integration of the informal collectors." The proximity of the dot to a stage (in the life cycle) and the line style represents the significance of the impact due to integration of informal collectors. Solid lines represent direct significant impact while dashed lines represent indirect impact which may not be proportionally too significant.

The third stage begins with the involvement of "kabadiwalas" who collect the e-waste generated when the second stage ends. Activities like extraction of valuable materials in the e-waste, dismantling of-waste and refurbishment of e-waste for further use are executed which pose occupational and environmental hazards. If the informal collectors will sell their collection to the formal recyclers, these activities will be done in a more regulated way, minimizing the negative impacts on health of people involved in the process and on the environment.

Hence, first stage is the entry of EEE. The informal collectors do not manufacture EEE but their collection could be refurbished and put to further use. The second stage is e-waste generation, where households and businesses use the EEE till it is of not much value to them. The informal collectors are themselves users of EEE, resulting in generation of e-waste.

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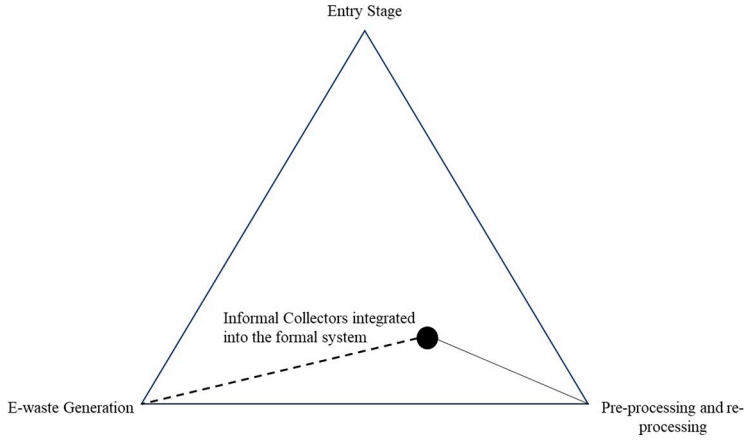


Figure 2. Integration of informal collectors in the e-waste management process
Source: Authors' own work

Thus, their integration along the life cycle of EEE could have an indirect impact (dotted lines) during the first and second stage.

7. Limitations and future research

The current research offers insights into some major challenges faced by the informal collectors in the e-waste management system in India and maps out interconnections between these challenges. However, the study has some limitations. First, data gathering relied on a sample of informal e-waste collectors and presents their understanding of the barriers related to formal recyclers and the regulators. A more comprehensive understanding to support the informal collectors' perceptions, necessitates additional insights from e-waste aggregators, certain formal e-waste collectors and regulators. Second, while the paper sheds light on barriers associated with government initiatives, the absence of interviews with government officials limits a holistic perspective. Third, to gain a more holistic perspective, insights from households are essential. Subsequent studies ought to explore the relationship between the tendency to hoard e-waste in Indian households and elements like concerns about data security, perceptions of value during disposal and the lack of awareness regarding formal e-waste recycling establishments and their procedures. Future studies may also include the objective of simulating the causal diagram through quantitative data collection, thereby reinforcing the findings proposed in this study and gauging the impacts of certain proposed policy decisions before they are put into force. To this end, the proposed causal diagram will require conversion to a stock and flow diagram and an understanding of the metrics to measure the subthemes included in the causal model.

Ethics statement

The research advisory committee as constituted by the coauthor's parent institution has duly approved the study conforming to ethical conduct. Respondents gave written consent before the interview.

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References

Akaka, M.A., Vargo, S.L., Nariswari, A. and O'Brien, M. (2021), "Microfoundations for macromarketing: a metatheoretical lens for bridging the Micro-Macro divide", *Journal of Macromarketing*, Vol. 43 No. 1, pp. 61-75.

Akter, N. and Hossain, M.I. (2025), "Structure, actors, and interdependencies between informal and formal e-waste management supply chain in Bangladesh", *Journal of Cleaner Production*, Vol. 522, p. 146146, doi: 10.1016/j.jclepro.2025.146146.

Alblooshi, B.G.K.M., Ahmad, S.Z., Hussain, M. and Singh, S.K. (2022), "Sustainable management of electronic waste: empirical evidences from a stakeholders' perspective", *Business Strategy and the Environment*, Vol. 31 No. 4, pp. 1856-1874, doi: 10.1002/bse.2987.

Andeobu, L., Wibowo, S. and Grandhi, S. (2021), "A systematic review of e-waste generation and environmental management of Asia pacific countries", *International Journal of Environmental Research and Public Health*, Vol. 18 No. 17, p. 9051, doi: 10.3390/ijerph18179051.

Andreasen, A.R. (1994), "Social marketing: its definition and domain", *Journal of Public Policy and Marketing*, Vol. 13 No. 1, pp. 108-114.

Ammirato, S., Linzalone, R. and Felicetti, A.M. (2022), "The value of system dynamics' diagrams for business model innovation", *Management Decision*, Vol. 60 No. 4, pp. 1056-1075.

Arya, S. and Kumar, S. (2020), "E-waste in India at a glance: current trends, regulations, challenges and management strategies", *Journal of Cleaner Production*, Vol. 271, p. 122707.

Biswas, A., and Singh, S. (2020), Centre for Science and Environment, New Delhi.

Borthakur, A. (2020), "Policy approaches on e-waste in the emerging economies: a review of the existing governance with special reference to India and South Africa", *Journal of Cleaner Production*, Vol. 252, p. 119885.

Braun, V., and Clarke, V. (2021), *Thematic Analysis: A Practical Guide*, SAGE Publications, London.

Bryant, A., and Charmaz, K. (2007), "Grounded theory in historical perspective: an epistemological account", In A. Bryant and K. Charmaz (Eds), *The SAGE Handbook of Grounded Theory*, SAGE Publications, London, pp. 31-57.

Byrne, D., and Callaghan, G. (2014), *Complexity Theory and the Social Sciences – The State of the Art*, Routledge, New York, NY, USA.

Corbin, J., and Strauss, A. (2008), *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*, Sage Publications, London.

Coyle, G. (2009), "Qualitative and quantitative modelling in system dynamics", *System Dynamics*, Vol. 2, p. 33.

Davis, J.M. and Garb, Y. (2015), "A model for partnering with the informal e-waste industry: rationale, principles and a case study", *Resources, Conservation and Recycling*, Vol. 105, pp. 73-83.

de Oliveira Neto, J.F., Souza, A.P.S., Silva, M.M., Santos, S.M. and Florencio, L. (2025), "Critical barriers to material recovery from e-waste in Brazil", *Journal of Hazardous Materials Advances*, Vol. 17, p. 100562, doi: 10.1016/j.hazadv.2024.100562.

Dhir, A., Koshta, N., Goyal, R.K., Sakashita, M. and Almotairi, M. (2021), "Behavioral reasoning theory (BRT) perspectives on e-waste recycling and management", *Journal of Cleaner Production*, Vol. 280 No. 1, p. 124269, doi: 10.1016/j.jclepro.2020.124269.

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Domegan, C. (2008), "Social marketing: implications for contemporary marketing practices classification scheme", *Journal of Business and Industrial Marketing*, Vol. 23 No. 2, pp. 135-141.

Domegan, C.M. and Calis, T. (2017), "Non-linear causal modeling in social marketing for wicked problems", *Journal of Social Marketing*, Vol. 7 No. 3, pp. 305-329.

Domegan, C., Brychkov, D., McHugh, P., McNamara, Á., Harkin, K., Fitzgerald, C. and O'Donovan, D. (2020), "Marketing systems: a listen, learn, leverage framework", *Journal of Macromarketing*, Vol. 40 No. 3, pp. 380-395.

Du, L., Feng, Y., Lu, W., Kong, L. and Yang, Z. (2020), "Evolutionary game analysis of stakeholders' decision-making behaviours in construction and demolition waste management", *Environmental Impact Assessment Review*, Vol. 84, p. 106408.

Duane, S. and Domegan, C. (2019), "Social marketing partnerships: evolution, scope and substance", *Marketing Theory*, Vol. 19 No. 2, pp. 169-193, doi: 10.1177/1470593118799810.

Duane, S., Duane, S., Domegan, C. and Bunting, B. (2022), "Partnering for UN SDG #17: a social marketing partnership model to scale up and accelerate change", *Journal of Social Marketing*, Vol. 12 No. 1, pp. 49-75, doi: 10.1108/JSOCM-10-2020-0200.

Dundon, T. and Ryan, P. (2010), "Interviewing reluctant respondents: strikes, henchmen, and gaelic games", *Organizational Research Methods*, Vol. 13 No. 3, pp. 562-581.

Dutta, D. and Goel, S. (2021), "Understanding the gap between formal and informal e-waste recycling facilities in India", *Waste Management*, Vol. 125, pp. 163-171.

Elias, A.A., Cavana, R.Y., and Jackson, L.S. (2001), "Stakeholder analysis to enrich the systems thinking and modelling methodology", *Proceedings of the 19th international conference of the systems dynamics society* (pp. 52-70).

Finn, B.M., Cobbinah, P.B. and Gounaridis, D. (2025), "The informal paradox: electronic waste and the toxic circular economy in Ghana", *Npj Urban Sustainability*, Vol. 5 No. 1, p. 101, doi: 10.1038/s42949-025-00299-5.

Forti, V., Balde, C., Kuehr, R., and Bel, G. (2020), *The Global e-Waste Monitor 2020: Quantities, Flows and the Circular Economy Potential*, United Nations Institute for Training and Research, International Telecommunication Union, and International Solid Waste Association, Bonn, Geneva and Rotterdam.

Freeman, R.E. (1984), *Strategic Management: A Stakeholder Approach*, Pitman, Boston.

Frey, B.S. (2000), "Motivation and human behaviour", *Risk, Trust and Welfare*, pp. 31-50.

Grandhi, S.P., Dagwar, P.P. and Dutta, D. (2024), "Policy pathways to sustainable e-waste management: a global review", *Journal of Hazardous Materials Advances*, Vol. 16, p. 100473, doi: 10.1016/j.hazadv.2024.100473.

Guo, S. and Chen, L. (2022), "Why is China struggling with waste classification? A stakeholder theory perspective", *Resources, Conservation and Recycling*, Vol. 183, p. 106312, doi: 10.1016/j.resconrec.2022.106312.

Gupta, P., Chauhan, S., Paul, J. and Jaiswal, M.P. (2020), "Social entrepreneurship research: a review and future research agenda", *Journal of Business Research*, Vol. 113, pp. 209-229.

Hennink, M.M., Kaiser, B.N. and Marconi, V.C. (2017), "Code saturation versus meaning saturation: how many interviews are enough? ", *Qualitative Health Research*, Vol. 27 No. 4, pp. 591-608, doi: 10.1177/1049732316665344.

Hodgkins, S., Rundle-Thiele, S., Knox, K. and Kim, J. (2019), "Utilising stakeholder theory for social marketing process evaluation in a food waste context", *Journal of Social Marketing*, Vol. 9 No. 3, pp. 270-287, doi: 10.1108/JSOCM-12-2017-0088.

Hurley, E., Dietrich, T. and Rundle-Thiele, S. (2019), "A systematic review of parent-based programs to prevent or reduce alcohol consumption in adolescents", *BMC Public Health*, Vol. 19 No. 1, p. 1451.

Downloaded from www.emerald.com/jsocm/article-pdf/16/3/427/11675805/jsocm-03-2025-0074en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA

Ibifunmilola, S.E. (2024), "Health and safety impact assessments among e-waste scavengers in Nigeria", (Doctoral dissertation, Manchester Metropolitan University).

Ikhlayel, M. (2018), "An integrated approach to establish e-waste management systems for developing countries", *Journal of Cleaner Production*, Vol. 170, pp. 119-130.

iSMA, ESMA, and AASM (2013), "Consensus definition of social marketing, 5th october 2013", available at: http://isma.memberclicks.net/assets/social_marketing_definition.pdf

Jain, M., Kumar, D., Chaudhary, J., Kumar, S., Sharma, S. and Verma, A.S. (2023), "Review on E-waste management and its impact on the environment and society", *Waste Management Bulletin*, Vol. 1 No. 3, pp. 34-44.

Kennedy, A. (2016), "Macro-social marketing", *Journal of Macromarketing*, Vol. 36 No. 3, pp. 354-365.

Kennedy, A., Kemper, J. and Parsons, A. (2018), "Upstream social marketing strategy", *Journal of Social Marketing*, Vol. 8 No. 3, pp. 258-279.

Kumar, A. (2021), "Electronic waste management in India", *Review. IITM Journal of Management and IT*, Vol. 12, pp. 19-23.

Kumar, S., Agarwal, N., Anand, S. and Rajak, B. (2022a), "E-waste management in India: a strategy for the attainment of SDGs 2030", *Materials Today: Proceedings*, Vol. 60 No. 2, pp. 811-814.

Kumar, A., Gaur, D., Liu, Y. and Sharma, D. (2022b), "Sustainable waste electrical and electronic equipment management guide in emerging economies context: a structural model approach", *Journal of Cleaner Production*, Vol. 336, p. 130391, doi: 10.1016/j.jclepro.2022.130391.

LeBel, S. (2012), "Wasting the future: the technological sublime, communications technologies, and e-waste", *Communication+*, Vol. 1 No. 1, pp. 1-19.

Lincoln, Y. and Guba, E.G. (1985), *Naturalistic Inquiry*, Sage, Newbury Park, CA.

Liu, H., Li, T. and Leong, G.K. (2022), "Choice of competitive strategy of formal and informal sectors in recycling WEEE with fund subsidies: service or price? ", *Journal of Cleaner Production*, Vol. 372, p. 133717, doi: 10.1016/j.jclepro.2022.133717.

Liu, H., Li, J., Wen, X. and Zhou, X. (2007), "Investigation on collection mode of domestic E-waste in urban China: the case of Beijing", *Proceedings of the 2007 IEEE International Symposium on IEEE*, pp. 257-261.

Luthra, A., Lowe, J. and Ochoa Berkley, K. (2023), "Dispossession by platformization: the rise of on-demand recycling in urban India", *Geoforum*, Vol. 141, p. 103731, doi: 10.1016/j.geoforum.2023.103731.

McKenzie-Mohr, D. and Schultz, P.W. (2014), "Choosing effective behavior change tools", *Social Marketing Quarterly*, Vol. 20 No. 1, pp. 35-46.

May, C. and Previte, J. (2016), "Understanding the midstream environment within a social change systems continuum", *Journal of Social Marketing*, Vol. 6 No. 3, pp. 258-276.

Nisha, B., Shajil, S., Dutta, R. and Jain, T. (2022), "Consumer awareness and perceptions about e-waste management in semi-urban area of Northern Tamil Nadu: a mixed-method approach", *Journal of Family and Community Medicine*, Vol. 29 No. 2, pp. 132-137, doi: 10.4103/jfcm.jfcm_318_21.

Padela, S.M.F., Wooliscroft, B. and Ganglmair-Wooliscroft, A. (2024), "Systematic theory mapping: deciphering causal complexity of brand externalities", *Journal of Macromarketing*, Vol. 44 No. 1, pp. 81-115.

Panwar, R., Ahmed, S. and Sharma, A. (2018), "Bridging the gaps between formal and informal e waste management in India with EPR", *International Journal of Development Research*, Vol. 8, pp. 18501-18506.

Panwar, R.M. and Ahmed, S. (2018), "Assessment of contamination of soil and groundwater due to e-waste handling", *Current Science*, Vol. 114 No. 1, pp. 166-173.

Downloaded from www.emerald.com/jsocm/article-pdf/16/3/427/11675805/jsocm-03-2025-0074en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA

Pejic Bach, M., Tustanovski, E., Ip, A.W., Yung, K.L. and Roblek, V. (2020), "System dynamics models for the simulation of sustainable urban development: a review and analysis and the stakeholder perspective", *Kybernetes*, Vol. 49 No. 2, pp. 460-504.

Radulovic, V. (2017), "Informal but integral", *E-Scrap News*, available at: https://resourcerecycling.com/e-crap/2017/09/18/informalutntegral/

Rautela, R., Arya, S., Vishwakarma, S., Lee, J., Kim, K.-H. and Kumar, S. (2021), "E-waste management and its effects on the environment and human health", *Science of The Total Environment*, Vol. 773, p. 145623.

Rebs, T., Thiel, D., Brandenburg, M. and Seuring, S. (2019), "Impacts of stakeholder influences and dynamic capabilities on the sustainability performance of supply chains: a system dynamics model", *Journal of Business Economics*, Vol. 89 No. 7, pp. 893-926.

Samuel, A. and Peattie, K. (2016), "Grounded theory as a macromarketing methodology: critical insights from researching the marketing dynamics of fairtrade towns", *Journal of Macromarketing*, Vol. 36 No. 1, pp. 11-26.

Sengupta, D., Ilankoon, I., Kang, K. and Chong, M. (2022a), "Circular economy and household e-waste management in India: integration of formal and informal sectors", *Minerals Engineering*, Vol. 184, p. 107661.

Sengupta, D., Ilankoon, I., Kang, K. and Chong, M. (2022b), "Circular economy and household e-waste management in India. Part II: a case study on informal e-waste collectors (kabadiwalas) in India", *Minerals Engineering*, Vol. 200, p. 108154.

Sharma, M., Joshi, S. and Govindan, K. (2021), "Issues and solutions of electronic waste urban mining for circular economy transition: an Indian context", *Journal of Environmental Management*, Vol. 290, p. 112373, doi: 10.1016/j.jenvman.2021.112373.

Shaw, D., Harris, F. and Ali, H. (2024), "Partnerships as strategy in Macro-Social marketing", *Journal of Macromarketing*, Vol. 44 No. 4, pp. 761-774, doi: 10.1177/02761467241267380.

Singhdeo, A.K., Tripathy, S. and Singhal, D. (2025), "Decoding e-waste challenges with hybrid AHP and ISM model approach: an initiative towards a cleaner future in India", *F1000Research*, Vol. 14, p. 856, doi: 10.12688/f1000research.167041.1.

Silva, S.L.D.A. and Sousa, F.L.D. (2018), "Towards the Brazilian space program modeling through the combination of stakeholder theory and system dynamics methodology", *Journal of Aerospace Technology and Management*, Vol. 10, p. e4818.

Singh, A., Panchal, R. and Naik, M. (2020), "Circular economy potential of e-waste collectors, dismantlers, and recyclers of Maharashtra: a case study", *Environmental Science and Pollution Research*, Vol. 27 No. 17, pp. 22081-22099.

Stepp, M.D., Winebrake, J.J., Hawker, J.S. and Skerlos, S.J. (2009), "Greenhouse gas mitigation policies and the transportation sector: the role of feedback effects on policy effectiveness", *Energy Policy*, Vol. 37 No. 7, pp. 2774-2787.

Sterman, J.D. (2000), *System Dynamics: Systems Thinking and Modeling for a Complex World*, McGraw-Hill.

Testa, S., Troise, C., Cincotti, S. and Camilleri, M.A. (2024), "The role of electronic waste management solutions and message framing in influencing consumer behaviours: exploring the crowdfunding context", *Business Strategy and the Environment*, Vol. 33 No. 2, pp. 917-929, doi: 10.1002/bse.3526.

Thukral, S. and Singh, M. (2023), "An exploratory study on producer's perspective towards e-waste management: a case of emerging markets", *Cleaner Waste Systems*, Vol. 5, p. 100090, doi: 10.1016/j.clwas.2023.100090.

Turaga, R., Bhaskar, K., Sinha, S., Hinchliffe, D., Hemkhaus, M., Arora, R., Chatterjee, S., Khetriwal, D.S., Radulovic, V., Singhal, P. and Sharma, H. (2019), "E-Waste management in India: issues and strategies", *Vikalpa: The Journal for Decision Makers*, Vol. 44 No. 3, pp. 127-162.

Downloaded from www.emerald.com/jsocm/article-pdf/16/3/427/11675805/jsocm-03-2025-0074en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA

Valai Ganesh, S., Suresh, V., Rajakarunakaran, S., Sathishkumar, P. and Rajkumar, M. (2025), "Sustainable electronic waste management framework for academic institutions in India", *Scientific Reports*, Vol. 15 No. 1, p. 40550, doi: 10.1038/s41598-025-24278-z.

Vanderby, S. and Carter, M. (2010), "An evaluation of the applicability of system dynamics to patient flow modelling", *Journal of the Operational Research Society*, Vol. 61 No. 11, pp. 1572-1581.

Wang, W., Tian, Y., Zhu, Q. and Zhong, Y. (2017), "Barriers for household e-waste collection in China: perspectives from formal collecting enterprises in Liaoning province", *Journal of Cleaner Production*, Vol. 153, pp. 299-308.

Wood, D.J., Mitchell, R.K., Agle, B.R. and Bryan, L.M. (2018), "Stakeholder identification and salience after 20 years: progress, problems, and prospects", *Business and Society*, Vol. 60 No. 1, pp. 196-245.

Wymer, W. (2011), "Developing more effective social marketing strategies", *Journal of Social Marketing*, Vol. 1 No. 1, pp. 17-31.

Wymer, W. (2021), "Addressing complex social problems with a multi-environmental stakeholder coalition", *International Review on Public and Nonprofit Marketing*, Vol. 18 No. 3, pp. 403-418, doi: 10.1007/s12208-021-00279-2.

Xi, Y., Rundle-Thiele, S., Sherring, P. and Roemer, C. (2026), "Waste not want not: a sequential social marketing pilot to reduce landfill destined e-waste", *Journal of Social Marketing*, Vol. 16 No. 1, pp. 58-71, doi: 10.1108/JSOCM-06-2025-0194.

Yadav, R., Panda, D.K. and Kumar, S. (2022), "Understanding the individuals' motivators and barriers of e-waste recycling: a mixed-method approach", *Journal of Environmental Management*, Vol. 324, p. 116303.

Yu, X., Sun, Z., Sun, D., Zhang, Y. and Li, M. (2025), "Sustainable development assessment of household e-waste reverse supply chains from an environmental ethic perspective", *Humanities and Social Sciences Communications*, Vol. 12 No. 1, p. 811, doi: 10.1057/s41599-025-05012-6.

Further reading

Aayog, N. (2023), *Mission Life*, Government of India, New Delhi.

Barton, S.L., Hill, N.C. and Sundaram, S. (1989), "An empirical test of stakeholder theory predictions of capital structure", *Financial Management*, Vol. 18 No. 1, pp. 36-44.

Bhardwaj, S.R. (2016), "International workshop on extended producer responsibility in India: opportunities, challenges and lessons from international experience", *Ministry of Environment, Forest and Climate Change, Government of India*, www.oecd.org/environment/waste/Session_1-EPR-in-E-waste-management-Indian-Prospective-Shruti_Rai.pdf

Borthakur, A. and Govind, M. (2018), "Public understandings of E-waste and its disposal in urban India: from a review towards a conceptual framework", *Journal of Cleaner Production*, Vol. 172, pp. 1053-1066.

Borthakur, A. and Sinha, K. (2013), "Electronic waste management in India: a stakeholder's perspective", *Electronic Green Journal*, Vol. 1 No. 36.

Chi, X., Streicher-Porte, M., Wang, M. and Reuter, M. (2011), "Informal electronic waste recycling: a sector review with special focus on China", *Waste Management*, Vol. 31 No. 4, pp. 731-742.

Conejo, F. and Wooliscroft, B. (2015), "Brands defined as semiotic marketing systems", *Journal of Macromarketing*, Vol. 35 No. 3, pp. 287-301.

Do, M., Nguyen, G., Thach, U., Lee, Y. and Bui, T. (2023), "Advances in hydrometallurgical approaches for gold recovery from E-waste: a comprehensive review and perspectives", *Minerals Engineering*, Vol. 191, p. 107977.

Gharaei, A., Karimi, M. and Shekarabi, A. (2019), "An integrated multi-product, multi-buyer supply chain under penalty, green, and quality control polices and a vendor managed inventory with consignment stock agreement: the outer approximation with equality relaxation and augmented penalty algorithm", *Applied Mathematical Modelling*, Vol. 69, pp. 223-254.

Downloaded from www.emerald.com/jsocm/article-pdf/16/3/427/11675805/jsocm-03-2025-0074en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA

Hawe, P., Shiell, A. and Riley, T. (2009), "Theorising interventions as events in systems", *American Journal of Community Psychology*, Vol. 43 Nos 3-4, pp. 267-276.

Hemkhaus, M., Henzler, M. and Hibler, S. (2026), "Enhancing resource efficiency through extended producer responsibility", Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH. European Union's Resource Efficiency Initiative (EU-REI) Project.

Hossain, M., M.Z.F., S, Al-Hamadani. and Rahman, M., (2015), "E-waste: a challenge for sustainable development", *Journal of Health and Pollution*, Vol. 5 No. 9, pp. 3-11.

Huff, A.D., Barnhart, M., McAlexander, B. and McAlexander, J. (2017), "Addressing the wicked problem of American gun violence: consumer interest groups as macro-social marketers", *Journal of Macromarketing*, Vol. 37 No. 4, pp. 393-408.

Hunt, S. (2020), "Institutional norms and the institutionalization of macromarketing: historical insights, the long macro view, and Service-Dominant logic", *Journal of Macromarketing*, Vol. 40 No. 3, pp. 286-293.

Joseph, K. (2006), "Stakeholder participation for sustainable waste management", *Habitat International*, Vol. 30 No. 4, pp. 863-871.

Kadirov, D. (2020), "Warmth rationing as a Macro-Social problem: the application of the chrematistics framework", In Kennedy, A. M. (Ed.), *Macro-Social Marketing Insights: Systems Thinking for Wicked Problems (*pp. 235-248). Routledge, New York, NY.

Kemper, J. and Ballantine, P. (2017), "Socio-Technical transitions and institutional change: addressing obesity through Macro-Social marketing", *Journal of Macromarketing*, Vol. 37 No. 4, pp. 381-392.

Kennedy, A. and Parsons, A. (2012), "Macro-social marketing and social engineering: a systems approach", *Journal of Social Marketing*, Vol. 2 No. 1, pp. 37-51.

Kumar, A. (2019), "Exploring young adults' e-waste recycling behaviour using an extended theory of planned behaviour model: a cross-cultural study", *Resources, Conservation and Recycling*, Vol. 141, pp. 378-389.

Kumar, A. and Dixit, G. (2018), "Evaluating critical barriers to implementation of WEEE management using DEMATEL approach", *Resources, Conservation and Recycling*, Vol. 131, pp. 101-121.

Liu, T., Cao, J., Wu, Y., Weng, Z., Senthil, R.A. and Yu, L. (2021), "Exploring influencing factors of WEEE social recycling behavior: a Chinese perspective", *Journal of Cleaner Production*, Vol. 312, p. 127829.

Nivedha, R. and Sutha, D. (2020), "The challenges of electronic waste (e-waste) management In India", *European Journal of Molecular and Clinical Medicine*, Vol. 7, pp. 4583-4588.

Ongondo, F., Williams, I. and Cherrett, T. (2011), "How are WEEE doing? A global review of the management of electrical and electronic wastes", *Waste Management*, Vol. 31 No. 4, pp. 714-730.

Panchal, R., Singh, A. and Diwan, H. (2021), "Economic potential of recycling e-waste in India and its impact on import of materials", *Resources Policy*, Vol. 74, p. 102264.

Raghupathy, D., Krüger, M., Chaturvedi, D., Arora, D. and Henzler, M. (2010), *Ewaste Recycling in India – Bridging the Gap between the Informal and Formal Sector. Government of India*, Ministry of Electronics and Information Technology (MeitY), New Delhi.

Ravindra, K. and Mor, S. (2019), "E-waste generation and management practices in Chandigarh, India and economic evaluation for sustainable recycling", *Journal of Cleaner Production*, Vol. 221, pp. 286-294.

Shultz, C.J., II, Rahtz, D. and Sirgy, J.M. (2017), "Distinguishing flourishing from distressed communities: Vulnerability, resilience, and a systemic framework to facilitate well-being", In R. Phillips and C. Wong (Eds), *The Handbook of Community Well-Being*, Springer, pp. 403-422.

Downloaded from www.emerald.com/jsocm/article-pdf/16/3/427/11675805/jsocm-03-2025-0074en.pdf?getftrtoken=AQECAHi208BE49Ooan9kkhW_Ercy7Dm3ZL_9Cf3qfKAc485ysgAAA

Shumon, M., Ahmed, S. and Islam, M. (2014), "Electronic waste: present status and future perspectives of sustainable management practices in Malaysia", *Environmental Earth Sciences*, Vol. 72 No. 7, pp. 2239-2249.

Tesfaye, F., Lindberg, D. and Hamuyuni, J. (2017), *Valuable Metals and Energy Recovery from Electronic Waste Streams*, In Z. e. al. (Ed) Springer, Cham.

Truong, V.D. (2017), "Government-led macro-social marketing programs in Vietnam: outcomes, challenges, and implications", *Journal of Macromarketing*, Vol. 37 No. 4, pp. 409-425.

Watson, F. and Ekici, A. (2017), "Well-being in alternative economies: the role of shared commitments in the context of a Spatially-Extended alternative food network", *Journal of Macromarketing*, Vol. 37 No. 2, pp. 206-216.

Wood, D.J., Mitchell, R.K., Agle, B.R. and Bryan, L.M. (2021), "Stakeholder identification and salience after 20 years: progress, problems, and prospects", *Business and Society*, Vol. 60 No. 1, pp. 196-245.

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